Biomedical Signal Processing and Control 88 (2024) 105607

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

Lesion identification in fundus images via convolutional neural network-vision transformer

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ARTICLE INFO

MSC:
0000
1111

Keywords:
Vision transformer
Classification
Retinal lesion
Deep learning

ABSTRACT

Purpose: Classification of fundus lesions plays a vital role in detecting some diseases in their early stages, including glaucoma and diabetes. During the clinical diagnosis and treatment prognosis, its higher accuracy will assist the doctors in optimizing the therapeutic schedule and reduce the workload of the ophthalmologists. However, the intrinsic characteristics of retinal lesions in retinal images make the detection process challenging.
Methods: In recent decades, many deep learning algorithms have been widely applied in various areas and achieved promising outcomes. Among the deep learning approaches, due to the overall success of Transformers in the natural language processing field, plenty of researchers have begun to explore the applicability of Transformer models in clinical applications such as recognizing various ophthalmic diseases. In this study, we propose a vision transformer-based pipeline for accurately classifying retinal diseases. To fully exploit the local and global associations between the individual image patches, a convolutional neural network concatenated with a vision transformer is employed to form the proposed framework. It can improve the performance of retinal lesion classification compared with a single-vision transformer or a convolutional neural network.
Results: Moreover, we pre-train our proposed framework on the ImageNet ISLVRC dataset and a sizeable retinal image database in sequence. Then, we fine-tune this vision transformer on downstream fundus image classification tasks.
Conclusion: Experimental results demonstrate that the proposed approach achieves superior performance on two publicly available datasets over state-of-the-art deep learning-based techniques, including convolutional neural networks and other vision transformers.

1. Introduction

The retina is one of the primary components of the eye, which supports the human visual system. It plays a role in converting the light entering the eye into electrical pulses that travel to the brain by the optical nerve. According to the nature of the retina, performing an accurate analysis of the retina is vital for ensuring the health and living standards of human lives [1]. First, the retina's anatomical changes can unveil the lesion caused by certain diseases. Meanwhile, Ophthalmic and systemic disorders, including diabetic retinopathy (DR), glaucoma, and age-related macular degeneration (AMD), are of great severity since they can result in partial or total loss of vision [2].

Nevertheless, an early diagnosis and screening instrument can contribute to avoiding or mitigating the considerable consequences. Accordingly, color retinography has become an affordable and widely accepted component in fundus imaging techniques, which can be exploited for early diagnosis and pathological screening. As shown in Fig. 1, three common retinal conditions are provided, including healthy, DR, and glaucoma. Accordingly, the classification of retinal images becomes a crucial task in clinical practice [3].

Automatic detection of fundus disease has become a necessary procedure to yield accurate outcomes for an early diagnosis, and timely treatment [2]. It is because manual analysis and diagnosis of ocular fundus diseases in retinal images are time-intensive and labor tedious. Moreover, manual inspection significantly depends on expertly trained ophthalmologists and manually delineated indicators, including microaneurysm, hemorrhage, disc size, and cup-to-disc ratio. Automated algorithms designed for retinal images can relieve the burden of ophthalmologists and physicians in the diagnosis and early treatment and

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https://doi.org/10.1016/j.bspc.2023.105607
Received 8 June 2023; Received in revised form 24 September 2023; Accepted 15 October 2023
Available online 19 October 2023

Fig. 1. Retinal lesions in fundus images from HRF dataset [4]. (Top) Health; (Middle) Diabetic retinopathy; (Bottom) Glaucoma.

decrease the variability in the interpretation of the same batch of images. Previously, plenty of machine learning-based algorithms has been broadly employed in retinal image analysis.

For instance, in the work of [5], the authors proposed an algorithm to identify the retinal blood vessels effectively. In their algorithm, the curvelet transform was exploited to implement the representation of the edges. Moreover, curvelet transform coefficients were adjusted to improve the segmentation by enhancing the edges in retinal images. Moreover, morphology operators with multi-structure were also employed to reveal the ridges in a retinal image further while simultaneously eliminating the unrelated ridges and preserving the thin vessels. Mansour et al. [6] presented an effective method for detecting and classifying hard exudates associated with DR in retinal images. The proposed classification algorithm is built upon the discrete cosine transform (DCT) and support vector machine (SVM) using color information. To accelerate the diagnosis of diabetic disease, Omar et al. [7] proposed a multi-label learning model to implement the automatic classification of exudate lesions. In this study, the multi-scale binary patterns (LBP) were used in feature extraction first. Meanwhile, the proposed approach consisted of a multi-label k-nearest neighbor (ML-KNN), a multi-label SVM, a multi-label neural network, and a multi-label neural network with back-propagation.

On the other hand, deep learning models can also be leveraged to improve the classification accuracy of related diseases. For instance, Arunkumar and Karthigaikumar et al. presented a feature extraction method based on deep learning for diagnosing retina. The difficulty of this task is that retina-based disease, such as age-related molecular degeneration and retinal detachment, shares common characteristics. Furthermore, the proposed deep learning model can deal with this issue by performing subtle feature extraction and reducing the feature dimensionality. Instead of using a manually-crafted and heuristic feature set, Li et al. [8] proposed a deep learning algorithm to perform automated feature extraction for identifying blood vessels in retinal images. In the work of [9], Muhammad et al. put forward a two-stage deep learning pipeline composed of two separate convolutional neural networks (CNN). The former CNN extracts the optic disc in a retinal image. Moreover, the latter CNN is then used to discriminate the generated optic discs into glaucomatous or healthy ones. To perform DR screening effectively, Murugan and Roy [10] proposed a CNN architecture to classify microaneurysms and non-microaneurysms image patches, which is widely taken as the initial sign of DR in clinical practice.

As the most common deep learning models, many CNN-based techniques have proved their superior performance in extracting discriminative representation of retinal images, which contributes to recognizing unnoticeable characteristics, over the traditional machine learning methods. However, they cannot learn the global associations between long-range pixels in retinal images since CNN adopts a local receptive field mechanism. During a convolution process, a local receptive field denotes the area occupied by an input exposed to a neuron within a convolutional layer.

To address the issues caused by the local receptive field, Vaswani et al. proposed the transformer architecture [11] that relies on the self-attention mechanism to unveil the long-range relations. Motivated by the first transformer employed in natural language processing (NLP), Desovitskiy et al. [12] presented the vision transformer to implement image classification. Due to its capability to capture the global association between long-range image patches, transformer models have achieved a promising outcome in machine vision tasks such as image classification.

For instance, in the work of [13], Lanchantin, Wang, and Qi presented a transformer for multi-label image classification. In contrast, the transformer is exploited to complicated dependencies between labels and image features. The proposed transformer employs a CNN to extract features from images as its input. A primary element of this method is the label mask using a ternary encoding to unveil the positive, negative, and unknown labels. Chen et al. [14] proposed a dual-channel transformer to generate powerful features by combining image patches and positional embeddings. The two channels address small and large patches in this transformer model.

Meanwhile, it leverages multiple attention modules to realize the complementation between channels. Moreover, a cross-attention-based fusion module using separate tokens from two channels is presented for exchanging information. Moreover, the given cross attention has a linear execution complexity. To classify a hyper-spectral image, which commonly contains sequential information from a group of spectra bearing long-range dependencies, He et al. [15] introduced a spatial–spectral transformer for hyperspectral image classification. Within this model, a CNN is exploited for spatial feature extraction, and a transformer is used to determine the association between sequential spectra. To note that He et al. pointed out that CNNs are incompetent to address sequential data due to their adoption of the local receptive field.

Another extensive application field of transformer models is medical image analysis. To alleviate the manual maneuver of pathologists caused by short of annotations, Wang et al. [16] proposed a hybrid algorithm that is pre-trained on unlabeled histopathological images in a self-supervised fashion. The proposed approach can reveal the histopathological images' inherent characteristics and domain-specific features by combining a CNN and a transformer. Ding et al. [17] proposed a transformer-based model that combines both CNN and transformer for pathology image classification. Specifically, the CNN is used to perform local feature extraction, and the transformer is exploited to excavate the global contextual information.

In general, the classification of fundus lesions is a vital mission to assist in diagnosing various retinal diseases. In this study, we propose a novel deep-learning framework consisting of a CNN followed by a vision transformer for classifying a set of lesions in retinal images to facilitate the diagnosis of DR. The main contributions of this work include the followings:

- To implement the automated retinal image analysis, we propose a CNN-vision transformer model to discriminate the lesions and non-lesions in retinal fundus images. To the best of our knowledge, this is an early work of a vision transformer-based algorithm in retinal image classification.
- We fully use the local and global associations between the patches in retinal images through the sequential CNN-vision transformer architecture.
- Experimental results demonstrate that the proposed approach has achieved superior performance over the state-of-the-art techniques.

The remaining article is organized as follows: Section 2 provides the details of the proposed approach, such as the structure of the CNN and transformer models. Section 3 presents the experimental design and outcome of the proposed method, as well as the comparison between our process and the state-of-the-art. Section 5 brings this study to a conclusion, including the advantage and disadvantages of the proposed algorithm.

2. Materials and methods

The presented approach is constructed upon a transformer combined with a CNN model. In the first phase, we introduced the deep CNN model Inception-Resnet-v2 [18] by removing the last three blocks (average pooling, dropout, and softmax) from the backbone. The output of the presented Inception-Resnet-v2 is exploited as the input for the subsequent vision transformer. We trained this pipeline on a publicly available retinal image dataset in the second phase. At last, we evaluated the well-trained model on several retinal image datasets. Fig. 2 illustrates the pipeline of our approach, and Fig. 3 provides the details of the stem module in the presented pipeline.

2.1. Feature extraction by Inception-Resnet-v2 backbone network

To provide a substantial input for the presented transformer, we introduced the CNN model Inception-Resnet-v2. It is composed of the inception modules and residual network units simultaneously. The primary hallmark of Inception is the optimized usage of inherent computation resources. Furthermore, the intuition of the Inception model initially proposed in the work of [19] is to augment both the width and depth of a network while maintaining the computational budget unchanged. Specifically, the Inception network can retain abundant features from multi-scaled convolutional kernels while significantly reducing the complexity of the model by decreasing the number of convolutional kernels. An incarnation of the Inception model named after GoogLeNet achieved state-of-the-art classification in the ImageNet Large-Scale Recognition Challenge 2014 (ILSVRC14). As shown in Figs. 4, 5, and 6, there are three inception modules in the proposed feature extractor in Fig. 2. In addition, these inception modules consist of convolutional units with different sizes of 1 × 1, 3 × 3, 5 × 5, 1 × 7, and 7 × 1. Specifically, the 1 × 1 convolution operator is used for dimension reduction. The merge-divide-merge structure of each inception module provides substantial support for unveiling the inherent representation.

Table 1
The settings of hyper-parameters used in this study.

Parameter	Setting
Batch size	8
Image resolution	299 × 299
No. of classes	5 or 2
optimizer	Adam
Learning rate	1e−4
Patch size	16
Depth	12
Epochs	100

Meanwhile, the first residual network raised in [20] was leveraged to train intense networks, e.g., GoogLeNet. The residual model can implement the bidirectional transmission between any two layers in the network. Moreover, it contributes to avoiding the disappearance of the gradient by skipping connections and converging in a faster and more manageable fashion. By treating the layers as learning residual functions referring to layer inputs, the residual learning framework won the championship in ILSVRC 2015. In general, the combination of Inception and residual modules can complement each other to enhance the extracted feature's validity and reduce the amount of computation. As shown in Figs. 7 and 8, there are two reduction blocks sandwiched between the three inception modules in Fig. 2.

In the introduced Inception-Resnet-v2 backbone, the combination of the inception module and residual module improves the classification accuracy and reduces the number of computations concurrently. Furthermore, we removed three modules from the Inception-Resnet-v2 model [18], including average pooling, dropout, and softmax, which are responsible for classification rather than feature extraction initially.

2.2. Retinal lesion classification by vision transformer

By replacing three modules with the presented transformer, the proposed model can take full advantage of the spatial information between long-range image patches. Meanwhile, we introduced the vision transformer shown in [12] in the proposed pipeline. As a pure transformer model, there are not any convolutional operators within this transformer.

In the first place, the original transformer [11] analyzes and processes NLP issues, while the introduced vision transformer takes the image as its input. To be compatible with the vision transformer, the output integrated feature map $X \in \mathbb{R}^{H\times W\times C}$ from the last layer of the feature extractor in Fig. 2 is divided into patches with the size of $P \times P$. Specifically, H, W, and C represent an image's height, width, and the number of channels. Furthermore, P is set to 16 since we follow the preferred setting of the study [12]. Accordingly, the proposed transformer has $N = \frac{HW}{P^2}$ patches for each input image. With the flattened image patches and linear embeddings of patches blocks (as shown in Fig. 2), the image patches can be flattened into 1 Dimension and represented into D dimensions sequentially. Along with the image patches, the 1 Dimension positional embedding is integrated to provide information on spatial distribution for the image patches. In addition, a learnable class token is attached to the image patches and position embeddings to represent the class of the entire input image. The image

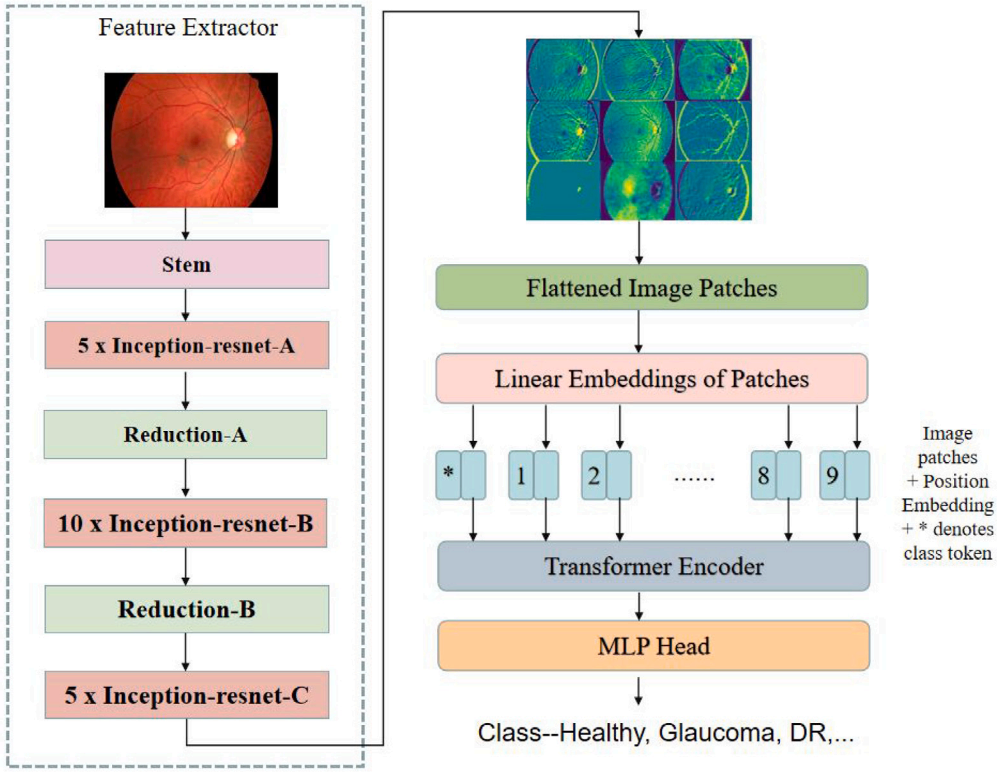

Fig. 2. Overview of the CNN-Transformer pipeline proposed for retinal lesion detection. The details of the modules used in this pipeline, including Stem, Inception-resnet-A, Inception-resnet-B, Inception-resnet-C, Reduction-A, Reduction-B, and Transformer Encoder, are provided in Figs. 3, 4, 5, 6, 7, 8, 9, respectively.

patches, position embeddings, and class token are then fed into the transformer encoder in the proposed pipeline (as shown in Fig. 2). In addition, Fig. 9 demonstrates the inner structure of the transformer encoder.

Precisely, the encoder consists of a multi-head self-attention (MSA) module, multi-layer perceptron (MLP), and normalization modules, which can mathematically be formulated as:

$$Z'_L = MSA(LayerNorm(Z_{L-1})) + Z_{L-1}, \tag{1}$$

$$Z_L = MLP(LayerNorm(Z'_L)) + Z'_L, \tag{2}$$

where L denotes the number of layers in the MLP module, and LayerNorm(.) represents the normalization operator [21]. Specifically, the MSA module follows the self-attention (SA) mechanism. And the MLP block consists of two linear layers, in which the first layer is in front of a Gaussian error linear unit (GELU).

2.3. Pretraining and fine-tuning

We pre-train the CNN-Transformer on large datasets to guarantee the proposed model's classification performance. We fine-tune it to retinal lesion detection tasks with retinal fundus images. Specifically, the weighing parameters in the proposed model were initialized by using a specific version of ImageNet [22] rather than adopting a random start strategy. Therefore, it can avoid the learned inductive bias and accelerate the convergence. Afterward, the pre-trained model was fine-tuned on the downstream retinal fundus images, e.g., EyePACS [23]. In general, both the introduced Inception-Resnet-v2 and vision transformer are subject to the supervision of the class token with a loss function. We introduced the cross entropy loss function for the proposed pipeline, as shown below.

$$Loss(y, y') = \sum_{i=1}^{C} y_i log(y'_i), \tag{3}$$

where y and y' denote the ground-truth label and prediction of the label, respectively. We can then obtain the proposed model from the format of the presented loss function.

3. Experimental results

3.1. Implementation details

To pre-train the combined pipeline of an Inception-Resnet-v2 and a vision transformer, we leveraged the extensive natural image database ImageNet ISLVRC2012. The adopted ImageNet database consists of more than 120 million natural images from 1000 categories. The pre-training is a multi-classification task with binary cross-entropy as the objective function. The entire dataset is divided into 80% for training and the remaining 20% for validation. To fit the requirement of Inception-Resnet-v2, the images from ImageNet ISLVRC2012 are resized into a resolution of 299 × 299 during the pre-training process. Note that our proposed model did not leverage the pre-trained weighting parameters as the original vision transformer [12].

After the pre-training procedure, we further fine-tuned the proposed architecture for the retinal fundus image classification task. One publicly available retinal fundus image dataset was leveraged to fine-tune the proposed pipeline's performance, which is the RFMiD dataset [24]. This dataset contains 3200 retinal images with 46 classes of conditions annotated through adjudicated consensus of two senior ophthalmologists. The introduced Inception-Resnet-v2 can address an input image with a resolution of 299 × 299. And all the hyper-parameters used in the proposed model are provided in Table 1. Meanwhile, the intermediate input module for the vision transformer can resize the output from the Inception-Resnet-v2 and divide it into image patches. Moreover, the corresponding position embedding for each image patch is interpolated to deal with a set of image patches. For the retinal image classification training, the cross entropy loss function supervises the feature extractor and classification module in Eq. (3).

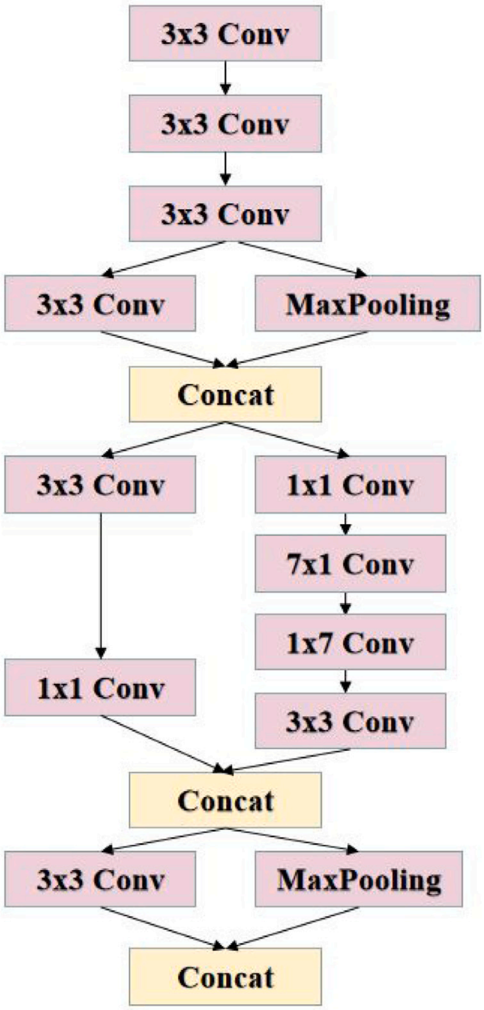

Fig. 3. Details of the stem module in the proposed pipeline.

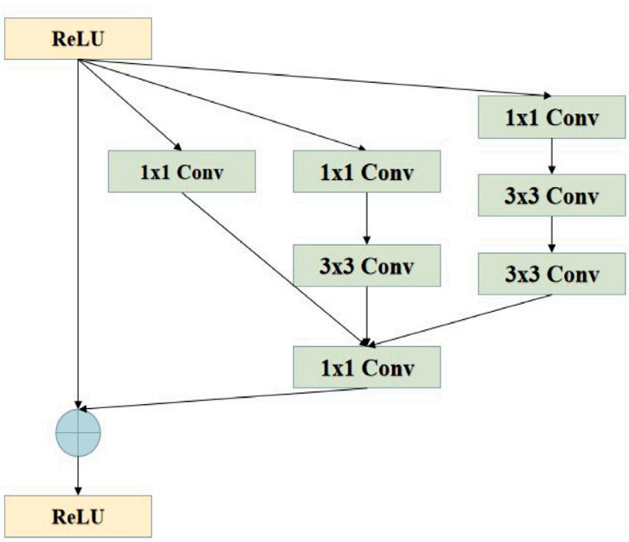

Fig. 4. Inception-resnet-A module in the proposed feature extractor.

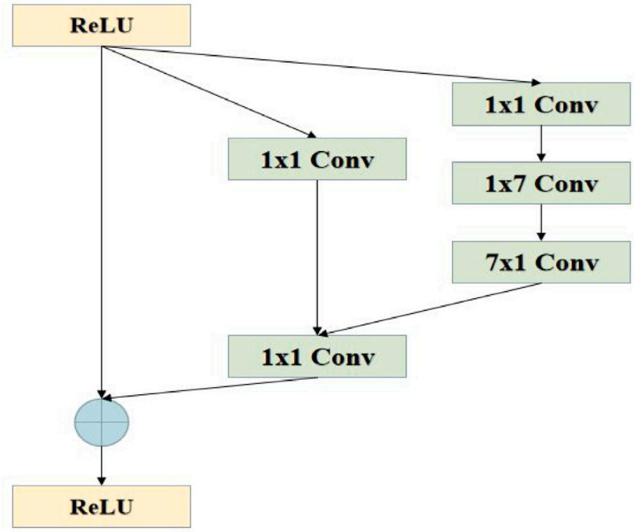

Fig. 5. Inception-resnet-B module in the proposed feature extractor.

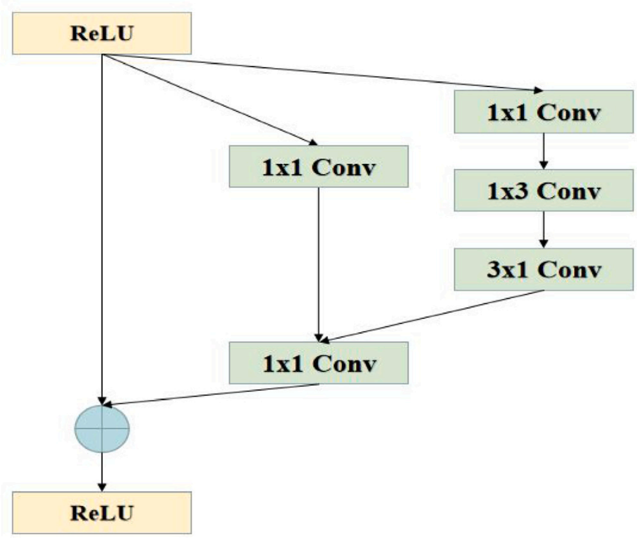

Fig. 6. Inception-resnet-C module in the proposed feature extractor.

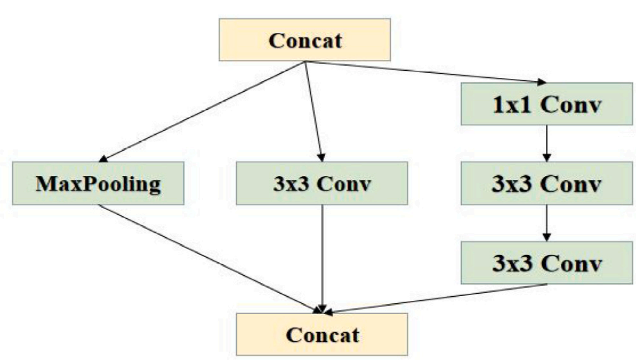

Fig. 7. Reduction-A module in the feature extractor.

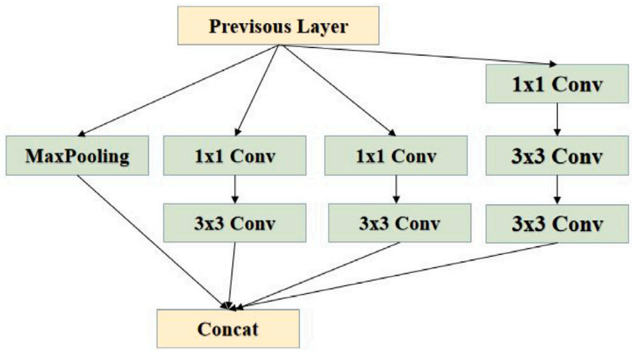

Fig. 8. Reduction-B module in the feature extractor.

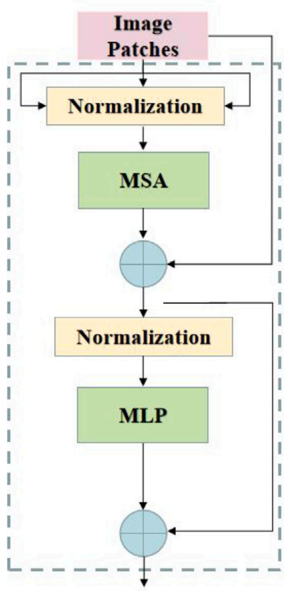

Fig. 9. The encoder in the presented vision transformer [12].

Table 2
The distribution of APTOS2019 and Messidor datasets.

Class (APTOS2019)	Number of images	Class (Messidor)	Number of images
DR 0	1805	DR 0	546
DR 1	370	DR 1	153
DR 2	999	DR 2	247
DR 3	193	DR 3	254
DR 4	295	–	–
Total	3662	–	1200

In the downstream fine-tuning phase, two publicly available datasets are used to evaluate the performance of the proposed approach, which includes the APTOS2019 database [25] and Messidor dataset [26]. In the former database, there are 3662 fundus images with five levels of DR grading from 0 (No DR) to 4 (severe), as shown in Table 2. For a fair comparison with the previous works [20,27–30], we leveraged 5-fold cross-validation to evaluate the effectiveness of the proposed model on APTOS2019. On the other hand, the Messidor dataset contains 1200 fundus images with DR grading information (as shown in Table 2) and diabetic macular edema (DME) labels. To compare with the previous works [28,31], We used images of DR at levels 0 and 1 from the Messidor while using a 10-fold cross-validation strategy for the DR 0 and DR 1 classification task. Furthermore, we leveraged the entire Messidor dataset for the DR grading task.

Table 3
Outcome of the ablation study on APTOS2019.

Combination		DR grading(%)			
Pre-train	Heads	AUC	Acc	wF1	wKappa
ImageNet ISLVRC	192	92.7	82.5	82.1	87.4
ImageNet ISLVRC	384	93.9	84.1	83.2	88.6
ImageNet +RFMiD	192	96.2	85.6	85.2	89.2
ImageNet+RFMiD	384	**98.3**	**89.1**	**87.8**	**91.8**

Table 4
Outcome of the ablation study on Messidor.

Combination		Disease detection(%)				
Pre-train	Heads	AUC	Acc	F1	Recall	Precision
ImageNet ISLVRC	192	91.2	91.8	90.0	88.4	91.6
ImageNet ISLVRC	384	93.6	92.7	92.0	91.7	92.4
ImageNet+RFMiD	192	94.5	93.2	93.9	93.3	94.5
ImageNet+RFMiD	384	**97.1**	**96.5**	**94.5**	**93.8**	**95.2**

Generally, the learning rate is set to 0.001, reduced by 0.5, each batch has eight images, and the maximum training epochs is 30. The data augmentation operations, including rotation, horizontal flipping, horizontal flipping, and cropping, are used in the training phase. The modeling and training are realized by PyTorch [32] with 4 NVIDIA Telsa V100 GPUs. The execution time of each image is 420 ms on average.

The evaluation metrics adopted in the experiments were the area under the curve (AUC), accuracy (Acc), Sensitivity, Specificity, Precision, Recall, and F1 score. Their mathematical expressions are formulated as follows:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN}, \tag{4}$$

$$Sensitivity = \frac{TP}{TP + FN}, \tag{5}$$

$$Specificity = \frac{TN}{TN + FP}. \tag{6}$$

$$Precision = \frac{TP}{TP + FP}, \tag{7}$$

$$Recall = \frac{TP}{TP + FN}, \tag{8}$$

$$F1 = 2 * \frac{Precision \times Recall}{Precision + Recall}, \tag{9}$$

where TP, TN, FP, and FN denote true positive, true negative, false positive, and false negative, respectively. Meanwhile, the weighted F1 score (wF1) and Kappa (wKappa) were also introduced as the evaluation metrics for the APTOS2019 dataset.

3.2. Ablation study

We conducted the ablation study to evaluate the performance of the proposed approach by pre-training on the ImageNet or the combination of ImageNet ISLVRC and RFMiD, respectively. Moreover, we evaluated the proposed model with different combinations of the embedding dimension (192 and 384) for the introduced vision transformer. In addition, the ablation studies were carried out on the APTOS2019 and Messidor datasets. For the multi-classification DR grading task in the APTOS2019 dataset, we adopted AUC, Acc, wF1, and wKappa. We used the metrics of AUC, Acc, F1 score, Recall, and Precision to classify the four categories of DR levels in the Messidor dataset. The outcome of the ablation studies is presented in Tables 3 and 4.

As demonstrated in Tables 3 and 4, for both the APTOS2019 and Messidor datasets, the pre-training on the combined datasets and the adoption of more heads in the MSA module improved the performance of the proposed approach under the evaluation metrics. For the DR

Table 5
Comparison between the state-of-the-art techniques and the proposed method on APTOS2019 (100%).

Model	AUC	Acc	wF1	wKappa
ResNet34 [20]	97.0	85.0	84.7	90.2
DLI [27]	–	82.5	80.3	89.5
CANeT [28]	–	83.2	81.3	90.0
GREEN-ResNet50 [29]	–	84.4	83.6	90.8
GREEN-ResNetNext50 [29]	–	85.7	85.2	91.2
MIL-VT [30]	97.9	85.5	85.3	**92.0**
Our methd	**98.3**	**89.1**	**87.8**	91.8

Table 6
Comparison between the state-of-the-art techniques and the proposed method on Messidor (100%).

Model	AUC	Acc	F1	Recall	Precision
Pires et al. [33]	86.3	–	–	–	–
CKML Net/LGI [34]	89.1	89.7	–	–	–
CANet [28]	89.5	81.0	–	–	–
Comprehensive CAD [35]	91.0	–	–	–	–
DSF-RFcara [36]	91.6	–	–	–	–
Expert [35]	94.0	–	–	–	–
Multi-task Net [37]	94.8	89.9	87.7	85.7	89.7
MTMR-Net [38]	94.9	90.3	88.3	86.7	90.0
Zoom-in-Net [39]	95.7	91.1	–	–	–
CANet + MultiTask [28]	96.3	92.6	91.3	92.0	90.6
SKD [31]	96.8	**96.9**	93.0	93.3	92.7
Proposed	**97.1**	96.5	**94.5**	**93.8**	**95.2**

grading task, the AUC, Acc, wF1, and wKappa with more datasets and more heads are enhanced by 6.04%, 8.00%, 6.94%, and 5.03%, respectively. And for the disease detection task, the AUC, Acc, F1 score, Recall, and Precision are improved by 6.47%, 5.12%, 5.00%, 6.11%, and 3.93%, respectively. The ablation studies' results suggest that the proposed pipeline's performance can be further enhanced by using more training data and fine-tuning its inner structure.

4. Discussion

4.1. DR grading

To compare the proposed framework with the state-of-the-art algorithms for DR grading, the corresponding comparison experiments were conducted on the APTOS2019 dataset. Meanwhile, the algorithms proposed in the works of ResNet34 [20], DLI [27], CANET [28], GREEN [29], and MIL-VT [30] were leveraged in the comparison experiments. To be specific, there are two versions of ResNet34 presented in [20], including the residual model trained on ImageNet and one residual network trained by a group of retinal fundus images. And GREEN [29] model also has two different architectures, which are GREEN-ResNet50 and GREEN-SE-ResNext50.

On the APTOS2019 dataset, as illustrated in Table 5, the proposed approach achieves superior performance over the state-of-the-art techniques except for one evaluation metric wKappa. To be specific, the MIL-VT [30] model has achieved better wKappa outcome than the proposed approach. However, the proposed approach is superior over the state-of-the-arts in terms of AUC, Acc, wF1 score, and wKappa.

Moreover, the confusion matrix for the DR grading task by using the proposed approach is provided in Fig. 10. As illustrated in Fig. 10, the proposed model can provide an overall Acc of 93.3% and a wKappa of 92.1% for the APTOS2019 dataset. Since there is an unequal distribution of DR grading images in the APTOS2019 dataset, the proposed approach has proven its potential value in dealing with the applications with imbalanced classification issues.

Note that the work of MIL-VT [30] has obtained a higher wKappa outcome on the APTOS2019 dataset than the proposed approach. However, a sizeable retinal fundus dataset was manually collected in the

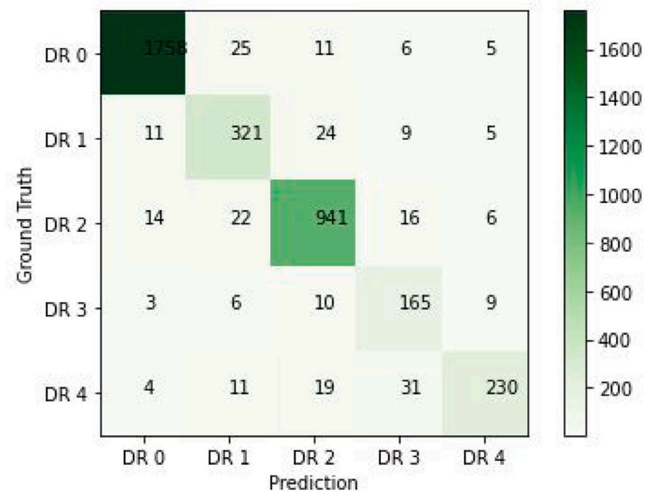

Fig. 10. The confusion matrix for DR grading on the APTOS2019 dataset by using the proposed approach.

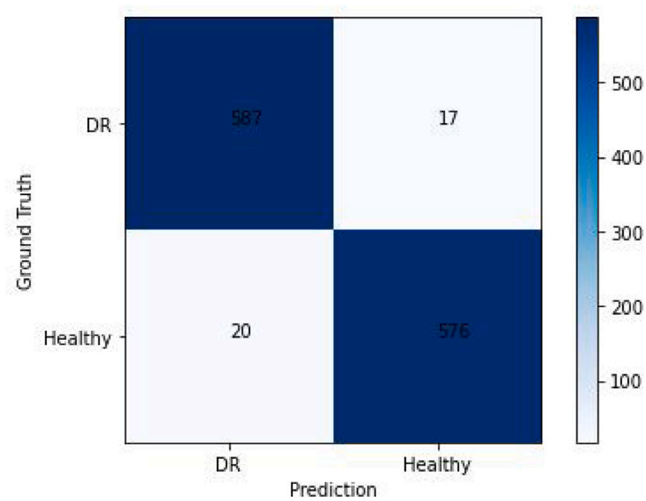

Fig. 11. The confusion matrix for DR and Healthy classification on the entire Messidor dataset.

study of MIL-VT [30]. This dataset contains 345,271 fundus images with five retinal conditions labeled and has a significant influence on the performance of MIL-VT, which has been proved by the experimental results in the work of MIL-VT. It reveals that the quantity of data samples has a significant influence on the performance deep learning models.

4.2. Lesion detection

In addition, to evaluate the performance of the proposed approach for lesion detection, we conducted comparison experiments between the state-of-the-art methods and the proposed approach for the binary classification of all samples in the Messidor dataset. It can be observed that SKD [31] has produced a superior performance in terms of Acc on the Messidor dataset over the proposed method. However, to note that the Acc of SKD is 92.9 ± 3.99 in the paper [31], and the sum of 92.9 and 3.99 is leveraged, which is the most optimal outcome of SKD, in Table 6. Meanwhile, our method has achieved superior performance over the state-of-the-art methods in terms of AUC, F1 score, Recall, Precision, simultaneously.

In addition, we illustrate the confusion matrix for the binary classification of all images in the Messidor dataset by using our approach.

Table 7
Comparison between the state-of-the-art techniques and the proposed method on Partial Messidor (100%) for binary classification (DR and Healthy).

Model	AUC	Acc	Sensitivity	Specificity
Expert A [35]	92.2	87.8	–	–
Expert B [35]	86.5	76.4	–	–
Holly et al. [40]	87.0	87.1	88.2	85.7
S^2MTS^2 [41]	86.3	86.7	88.7	84.8
SRC-MT [42]	84.8	85.8	86.4	85.2
ACCN [43]	96.0	89.8	93.0	86.7
Odena et al. [44]	96.7	94.7	95.4	**95.1**
Proposed	**97.1**	**95.3**	**96.6**	94.2

As shown in Fig. 11, the proposed pipeline correctly identifies 587 images with lesions and 576 healthy images with an Acc of 96.9%. For a fair comparison with the state-of-the-art algorithms [35,40–44], the comparison experiments for the binary classification of partial images in Messidor dataset were conducted. Half of the Messidor dataset was chosen as the evaluation set, with 300 healthy (DR 0) images and 300 DR (DR 1, DR 2, and DR 3) images. As shown in Table 7, the proposed approach achieves an outstanding outcome in terms of AUC, Acc, and Sensitivity. Note that the Specificity generated by the study Odena [44] is 95.1%, which is higher that the Specificity outcome of the proposed method 94.2%.

Besides, the confusion matrix for the binary classification of 600 images in the Messidor dataset by our approach is also provided. As shown in Fig. 12, the proposed pipeline correctly identifies 282 images with lesions and 290 healthy images with an Acc of 95.3%.

Different from the work of [30], which introduced the multi-instance learning (MIL) [45] into the vision transformer [12], the input of the transformer is replaced rather than modifying the output of this model. We put forward that the original vision transformer only uses the features connected to the class token while neglecting the components extracted from individual image patches. Specifically, each single image patch still contains relative information that can improve classification accuracy. Moreover, the class token with the feature representation leveraged in the original vision transformer requires complementary information from the image patches. For instance, the lesions may appear anywhere in a retinal image; thus, every patch should greatly value lesion detection in clinical practice. Note that the transformers were designed for dealing with sequential data. And each image patch in our proposed pipeline still contains position information of the entire image. By using the MIL mechanism, this work [30] adopts the MIL embedding, and an attention aggregation function incorporates the individual image patches.

Generally, the experimental results can be credited to that the pipeline composed of Inception-Resnet-v2 and a vision transformer can capture the feature embeddings locally and discover the associations globally. Meanwhile, since the last three layers from the original Inception-Resnet v2 model have been removed in the proposed pipeline, the extracted features can be further refined. By using the above-mentioned modules, the proposed approach could transform every retinal image into a cascaded feature map for retinal lesion detection. In addition, the cropped model structure in this study can be attached to other versions of vision transformers in a plug-and-play fashion.

Meanwhile, this study still has several limitations. First of all, the data samples used in this study is not enough in quantity. Since the quantity of images significantly relates to the outcomes of the deep learning models, this issue might have severely reduce the performance of the proposed approach. Secondly, the number of evaluation metrics used in this study is not sufficient. And more evaluation metrics can be incorporated into the experiments. Finally, the interior modules used in the proposed model should be optimized to generate a better outcome.

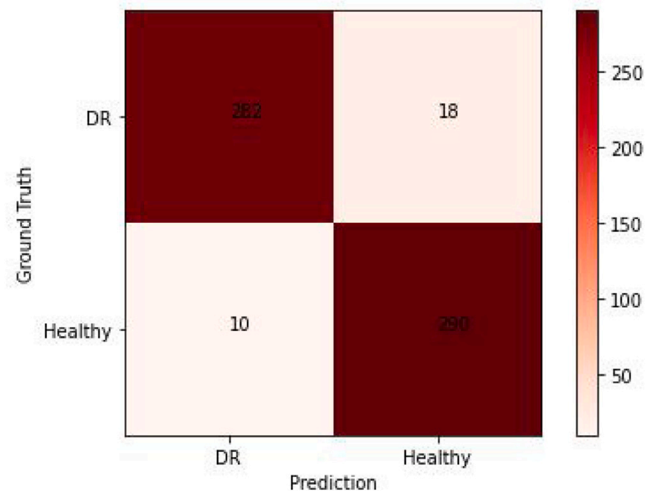

Fig. 12. The confusion matrix for DR and Healthy classification on the 50% Messidor dataset.

5. Conclusion

To address the retinal image classification issue, we leverage the Inception-Resnet-v2 to unveil multi-scale features from the input retinal image. Consequently, the input of our introduced vision transformer contains the feature embeddings from different scales of one single image. Using the resize operations, the input combination of multi-scale features can be divided into separate feature maps. Moreover, each patch is a feature map extracted from the Inception-Resnet-v2 model. Accordingly, the proposed approach fulfills the requirements, including features of the individual image patches and the corresponding positional distribution.

This paper explored the capability of combining CNN and vision transformers for retinal lesion classification. CNN architectures have achieved breakthrough image classification in the last few decades. And these networks can unveil the inherent characteristics of the images. However, they focus on the local receptive field within every single image. On the other hand, the vision transformer models can provide a global angle of view based on the feature embeddings extracted by using CNNs. With pre-training on a sizeable natural image database and fine-tuning on the retinal image datasets, the conjunctive model has proved its ability in retinal lesion detection and achieved superior performance over the state-of-the-art CNNs and vision transformers.

Future work includes combining other types of backbone networks as the feature extractor module and using various classifiers. Since it is a promising outcome for retinal image analysis, we will adapt the proposed architecture to classify other images.

CRediT authorship contribution statement

Jian Lian: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Funding acquisition. **Tianyu Liu:** Software, Validation, Writing – original draft, Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare no potential conflicts of interest.

Data availability

Data will be made available on request.

Acknowledgments

This work was made possible through support from the Natural Science Foundation of Shandong Province in China (No. ZR2020MF133) and the special fund for scientific research and cultivation of Shandong Police College, China (No. YKYPYZX202201).

Ethical approval and consent to participate

Approved.

Consent for publication

Approved.

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